

Oasis 3D Printer

Operations Report & Standard Operating Procedure (SOP)

Document Version: 1.0 | Date: 2026-03-31

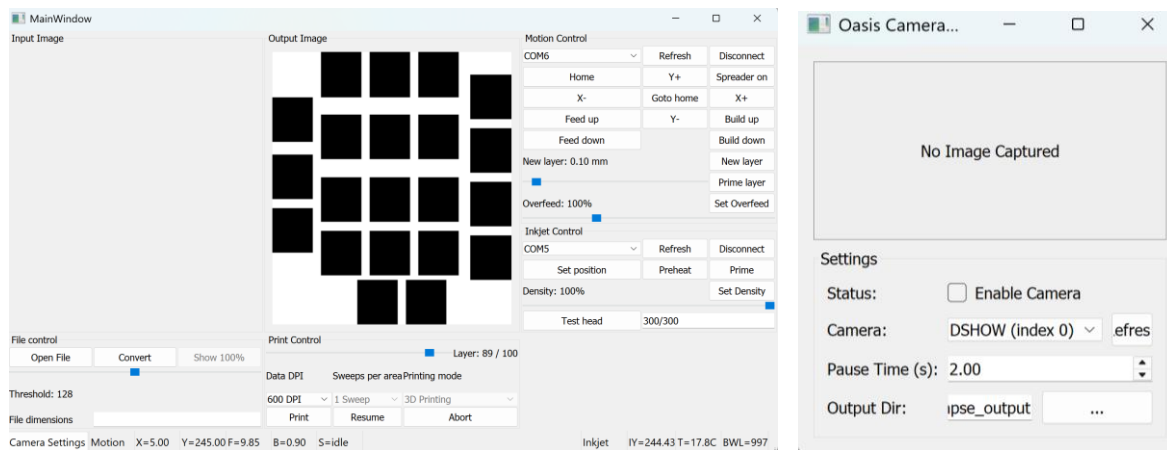
Software: Oasis_printer_260327.py + Study_params_loader_and_print.py

1. System Overview

1.1 Oasis_printer_260327.py

A PyQt5-based GUI application that simultaneously controls an HP45 inkjet printhead and a GRBL motion controller, implementing Binder Jetting 3D printing onto ceramic powder layers.

Component	Description
MainWindow	Main GUI. Split into self.ui (separate window from .ui file) and self.form (widget references)
CameraController	Per-layer image capture (separate widget)
GRBL	Motion serial communication (XY + Z/A rollers)
HP45	Inkjet printhead serial communication
ImageConverter	SVG/image → print array conversion



Example image for MainWindow.UI

1.2 Study_params_loader_and_print.py

An experiment automation script that reads parameters from oasis_change_parameters.csv and automatically injects them into the Oasis GUI. Guarantees a reproducible parameter set per run and saves execution records to the runs/ directory.

2. SOP — Standard Operating Procedure

2.1 Pre-Run Checklist

Step	Check Item
1	Review parameter values in oasis_change_parameters.csv (especially variance entries)
2	Confirm GRBL motion controller COM port (Device Manager)
3	Confirm HP45 inkjet controller COM port
4	Visually inspect powder fill level and printhead condition

3.2 GUI Startup Sequence (Critical)

⚠ The following sequence must be followed strictly. Deviating from this order is not supported.

Step	Action	Detail
1	Load SVG File (First Priority)	File → Open → Select the target SVG file. Must be done BEFORE connecting motor or printhead.
2	Connect Motor COM Port	Motion tab → Select COM port → Click Connect
3	Run Homing	Click Home button — confirm machine returns to origin position
4	Connect Printhead COM Port	Inkjet tab → Select COM port → Click Connect
5	Verify Temperature Reading	Confirm T (temperature) value appears in the bottom-right of the UI — indicates successful printhead connection

3.3 Printhead Nozzle Check & Priming Procedure

Perform the following procedure before every print to ensure print quality.

Step	Procedure
Step 1	Nozzle Test Click Test Nozzle → Verify all 300 nozzles fire without blockage <i>If any nozzles are blocked, proceed to Step 4 (Cleaning)</i>
Step 2	Preheat — 10 clicks Click Preheat and wait for T (temperature) to increase before clicking again Maintain ~1 second interval between clicks ⚠ Rapid repeated clicking will cause an error — always wait for temperature change before next click
Step 3	Prime — 5 clicks × 2 sets Click Prime 5 times → wait briefly → repeat once more (2 sets total) Visually confirm binder is being ejected from the printhead

Step 4	Cleaning Procedure (if poor jetting is observed) 1) Sonicate printhead in IPA (isopropyl alcohol) for 1 minute 2) Remove all moisture using a fine-cleaner (lint-free wiper recommended) 3) Reinstall the printhead 4) Repeat Steps 2–3 (Preheat + Prime) and re-verify jetting performance
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3.4 Parameter Configuration (oasis_change_parameters.csv)

Parameter	Type	Default	Unit	Range	Description
print_speed	variance	2200	mm/min	100–10000	Inkjet sweep speed
dpi	variance	300	dpi	100–1200	Print resolution
layer_passes	variance	3	—	1–5	Repeat passes per layer
density	variance	500	permille	0–1000	Ink firing density
layer_thickness	variance	0.1	mm	0.05–2	Layer thickness
overfeed_percent	variance	100	%	175–500	Default (100) is outside valid range — update to ≥ 175
dwelt_time	fixed	5	s	—	Wait time before camera capture

3.5 Experiment Record Management

Each run automatically saves to runs/<RUN_ID>/:

File	Contents
manifest.json	Timestamp, git hash, source CSV path, full resolved config
resolved_params.csv	Original CSV values vs. actually applied values (resolved_value column)

To reproduce an experiment: copy the resolved_value column from resolved_params.csv into the value column of oasis_change_parameters.csv and re-run.